

Zhuo Cai 📞 (+86) 17755650269 (+852) 67397012
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🌐 <https://zhuocai.github.io>

Education

- July – Oct 2025 📌 **University of Oxford, UK**
Visiting Scholar
- 2023 – Now 📌 **Hong Kong University of Science and Technology, Hong Kong**
PhD in Computer Science and Engineering
Co-Supervised by Amir Goharshady & Dimitris Papadopoulos
Research Interest: Cryptography, Distributed Systems, Formal Verification, Mechanism Design, Theoretical Computer Science.
- 2021 – 2023 📌 **Hong Kong University of Science and Technology, Hong Kong**
Master of Philosophy in Computer Science and Engineering
Supervisor: Amir Goharshady
GPA: 3.88/4.0
- 2017 – 2021 📌 **Tsinghua University, Beijing**
Bachelor of Automation
GPA: 3.81/4, GPA Ranking: 17/168
- 2019 Fall 📌 **National University of Singapore, Singapore**
Visiting Undergraduate Researcher
GPA: 4.0/4

Research Publications

Note: (1) My advisor Amir Goharshady adopts the convention in theoretical computer science to order authors alphabetically. (2) ‘*’ indicates a co-first author.

Publications using Alphabetical Order

1. Barakbayeva, T., Cai, X., **Cai, Z.**, & Goharshady, A. K. (2026). Extending Programmable Blockchain Protocols to Enable and Incentivize Parallel Execution of Smart Contracts. In MARBLE.
2. **Cai, Z.**, & Goharshady, A. (2026). Proof of Election: A Formally-Verified Democratic Blockchain Protocol. In IEEE DAPPS.
3. Ballweg, J., Barakbayeva, T., **Cai, Z.**, & Goharshady, A. (2026). LockFee: Refundable Transaction Admission via Time-Locked Collateral. In IEEE DAPPS.
4. **Cai, Z.**, & Goharshady, A. (2026). Brief Announcement: Delay-Optimal Transaction Order Fairness. In PODC. [full version](#)
5. Barakbayeva, T., **Cai, Z.**, Goharshady, A. & Key Poor K. (2025). Smart Contracts for Trustless Sampling of Correlated Equilibria. In IJCAI. [code - SNARK part](#)
6. Abidha V., Barakbayeva T., **Cai, Z.**, & Goharshady, A. (2024). Gas-efficient decentralized random beacons. In IEEE ICBC.
7. Barakbayeva T., **Cai, Z.**, & Goharshady, A. (2024). SRNG: an efficient decentralized approach for secret random number generation. In IEEE ICBC.

8. Cai, Z., Farokhnia, S., Goharshady, A., & Hitarth, S. (2023). Asparagus: Automated synthesis of parametric gas upper-bounds for smart contracts. In OOPSLA. [Asparagus code](#)
9. Ballweg, J., Cai, Z., & Goharshady, A. (2023). PureLottery: Fair leader election without decentralized random number generation. In IEEE Blockchain. [PureLottery code](#)
10. Cai, Z., & Goharshady, A. (2023). Trustless and bias-resistant game-theoretic distributed randomness. In IEEE ICBC.
11. Cai, Z., & Goharshady, A. (2023). Game-theoretic Randomness for Proof-of-Stake. In MARBLE.

Publications using Contribution Order

1. Pappas, C., Cai, Z., & Papadopoulos, D. (2026). Rogue: Updatable Matrix Lookup Arguments and Applications to Verifiable Databases. In CCS.
2. He, X.*, Cai, Z. *, Wei, W., Zhang, Y., Mou, L., Xing, E., & Xie, P. (2021). Towards visual question answering on pathology images. In ACL.

Miscellaneous Experience

Awards and Achievements

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| 2023 | <ul style="list-style-type: none"> 📌 Hong Kong PhD Fellowship 📌 Young Researcher, 10th Heidelberg Laureate Forum 📌 Research Travel Grant, HKUST |
| 2019 | <ul style="list-style-type: none"> 📌 Honor of Academic Excellency, Tsinghua University.
Awarded to top 10% students. |
| 2018 | <ul style="list-style-type: none"> 📌 Honor of Academic Excellency, Tsinghua University. |
| 2016 | <ul style="list-style-type: none"> 📌 1st Level in National High School Mathematics League, the Chinese Mathematical Society.
Ranked within the top 2% in the province of Anhui. |

Academic Service

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| 2026 | <ul style="list-style-type: none"> 📌 Technical Program Committee Member, IEEE BCCA'2026. 📌 Program Committee Member, MARBLE'2026. 📌 Reviewer, Journal BCRA'2026 Blockchain: Research and Applications. |
| 2025 | <ul style="list-style-type: none"> 📌 Program Committee Member, MARBLE'2025. |

Extracurricular Experience

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| Dec 2019 - May 2021 | <ul style="list-style-type: none"> 📌 President of iOS Club, Tsinghua University. |
| Aug 2018 - Aug 2019 | <ul style="list-style-type: none"> 📌 Leader of the Students' Association of Science and Technology (Competition Branch), Department of Automation, Tsinghua University. |

Teaching

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| 2026 Spring, Instructor | <ul style="list-style-type: none"> 📌 ALPACAS Research Group
Modern Blockchain Consensus |
| 2025 Spring & 2026 Spring, TA | <ul style="list-style-type: none"> 📌 Hong Kong University of Science and Technology
COMP5631: Cryptography and Security |

Teaching (continued)

2024 Spring, TA	📖	Hong Kong University of Science and Technology COMP 4541: Blockchains, Cryptocurrencies and Smart Contracts
2022 Spring & Fall, TA	📖	Hong Kong University of Science and Technology COMP 2012: Object-Oriented Programming and Data Structures

Skills

Languages	📖	Native in Mandarin Chinese, proficient in English (TOEFL:106, GRE:158+170).
Coding	📖	C/C++, Python, MATLAB, JAVA, PyTorch, HTML, $\text{\LaTeX}$, ...